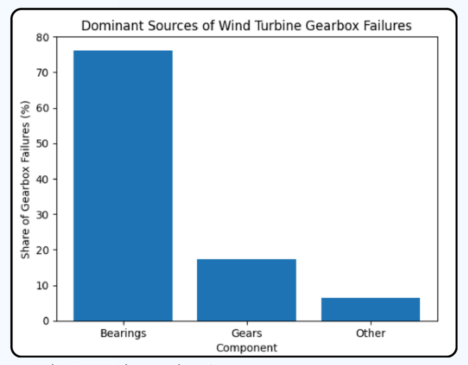
MAGLEV TURBINE: USING A HYBRID AMB-PERMANENT MAGNET SYSTEM FOR ROTOR LEVITATION TO ACHIEVE 0 FRICTION

A next-generation wind turbine with near-zero friction and minimal wear

PROBLEM

Wind turbine Bearings cause:

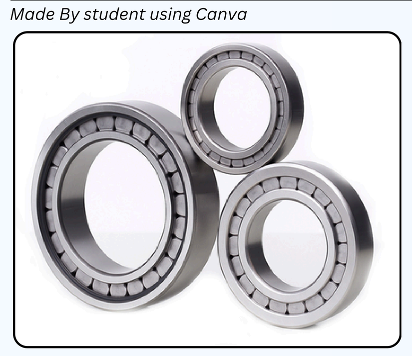
- **Continuous damage:** They cause continuous friction, generate heat, and need lubricant replacement
- **Failures**: 76% of recorded gearbox failures involve bearing damage



Made By student using Canva

End effects of Bearings:

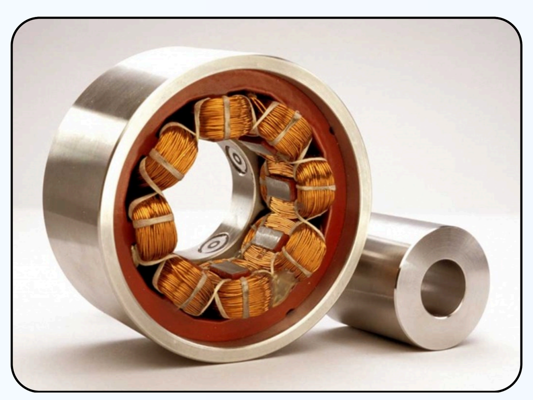
- **Maintenance costs:** Maintenance for these issues costs approximately 20–25% of the overall cost for modern wind turbine projects
- **Downtime**: Maintenance related downtime also causes these windmills to be out of operation for long periods
- **Lifetime reduction:** These bearings can also limit overall lifetime of windmills, due to accumulated fatigue.


Richard. (2025, August 18). The Advantages And Disadvantages Of Full Complement Bearings. Lily-Bearing.com; Shanghai Lily Bearing Limited.

Contact bearings are the central limitation of wind turbine designs, making projects more financially fragile over time. Reducing or eliminating mechanical contact thus represents a opportunity to improve long-term reliability and lifetime energy production.

SOLUTION INTRODUCTION

This project introduces a **hybrid Active Magnetic Bearing (AMB) wind turbine,** that uses a combination of stable and adjustable **magnetic forces** instead of conventional bearings. This hybrid approach reduces the power required for active control while eliminating continuous metal-to-metal contact.


admin. (2023, July 17). Innovative Bearing Technology: Non-Contact Magnetic Solutions. Pibsales.

Benefits of This Approach

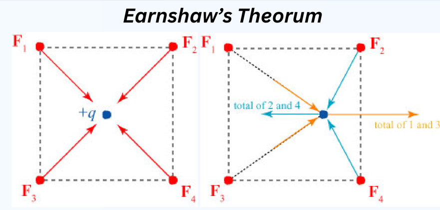
- Eliminates primary sources of bearing wear and failure
- Reduces maintenance requirements and downtime
- Increases long-term reliability and lifetime energy production
- Particularly advantageous in offshore or hard-to-access environments

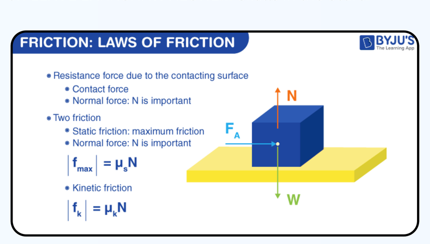

Hartsfield, T. (n.d.). Explainer: what is a Tesla coil? The Conversation. Retrieved February

CRITERIA FOR SUCCESS

- **Minimize Friction:** Reduce mechanical contact at the rotor shaft
- **Stability**: Maintain stable rotor operation with minimal vibrations, gyroscopic/precessional oscillations, or whirl modes.
- **Energy input**: Require low power input for active stabilization
- **Efficiency**: Demonstrate energy efficiency improvement
- **Low Stress**: Minimize thermal and vibrational stress, allowing for long-term reliability.
- **Backup Systems**: Ensure rotor safety during system disturbances, such as sudden loss of active coil power.

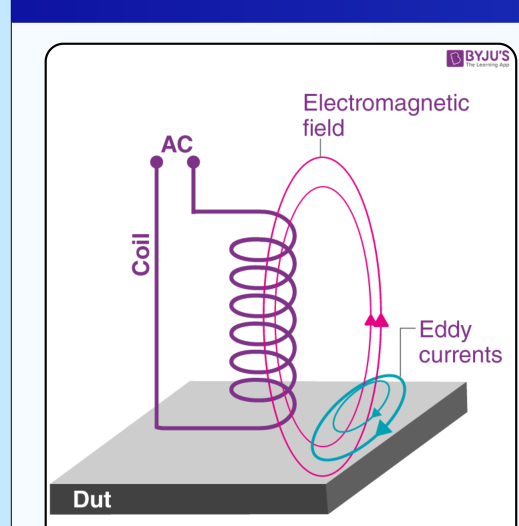
Earnshaw's Theorum


Thesat, S. (2009, April 14). Levitation and diamagnetism, or: LEAVE EARNSHAW ALONE!!! Skulls in the Stars.



Dr. Sasidharan Murugan. (2025, June 19). Gyroscopic Effect – The Hidden Force in Rotation! Linkedin.com.

CONSTRAINTS & SAFETY



What is Eddy Current? - Definition, Eddy Current Loss, Applications. (n.d.). BYJUS. Retrieved February 24, 2026

Design Constraints

- **Eddy drag minimization:** Rotor structures must avoid conductive materials near magnetic fields to prevent eddy current drag and heating.
- **Temperatures**: Temperature should stay low to prevent loss of magnetic strength or irreversible demagnetization.
- **Rotor displacement**: air gap must be controlled carefully, as small misalignments can cause rotor touchdown or instability.
- **Sensors**: Sensor systems must detect rotor movement accurately to maintain levitation and prevent instability.

WIND TURBINE PROTOTYPE DESIGN & ENGINEERING

Internal Rotor components

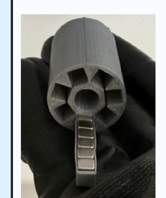


Image taken by finalist

Design

- Cylindrical rotor contains 7-sided internal magnet housing geometry
- All magnets oriented North pole outward.

We are using two lengths of arrays

- A 6 magnet rectangular prism array
- A 24 magnet prism array

This allows for the repulsion from the shorter AMB system and longer Halbatch while still making maximal use of resources

This contains the magnets, and is inserted into the cylinder

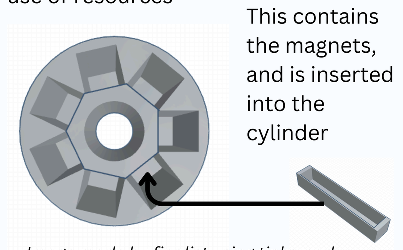


Image made by finalist using tinkercad

MAGLEV TURBINE SYSTEM OVERVIEW

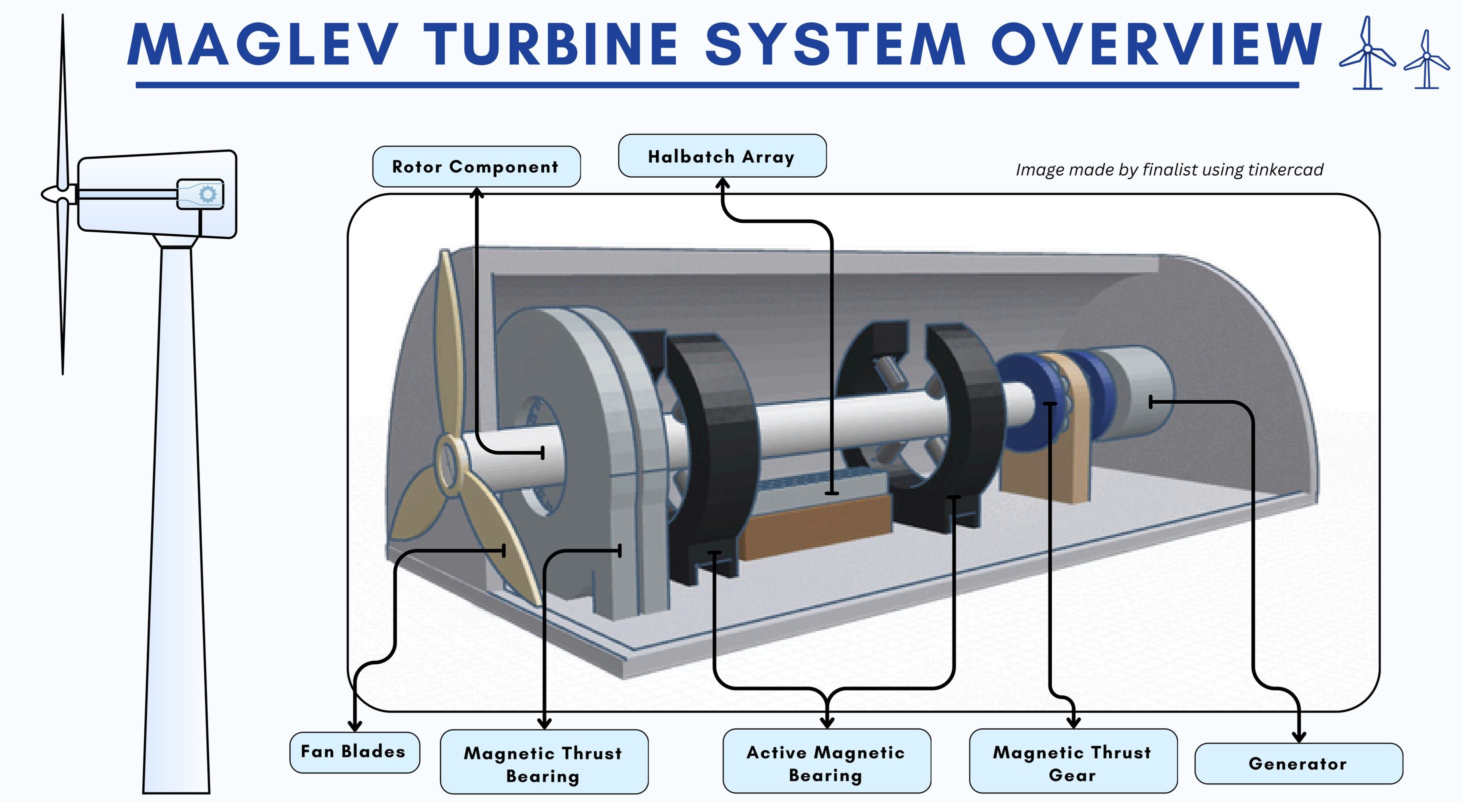
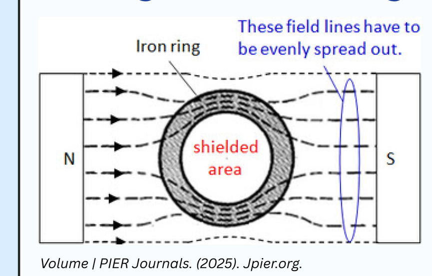


Image made by finalist using tinkercad

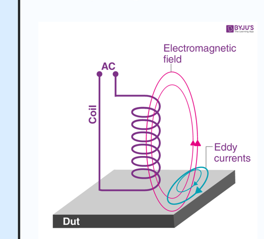
Image made by finalist using Canva

Sheilding & Magnetic Drag



Volume | PIER Journals. (2025). Jpier.org.

Soft magnetic shielding encloses the generator to confine stray flux and protect electronics without disrupting system fields.


Electromagnets Explained: Types, Materials, Working Principles, and Key Applications.

Non-conductive structural materials eliminate eddy current formation, minimizing parasitic drag and thermal losses.

Magnetic Flux Gear

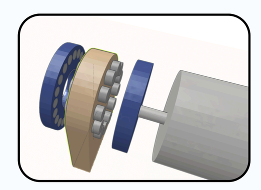


Image made by finalist using tinkercad

Magnetic flux lines from input magnets are guided through the ferromagnetic array and interact with output magnets. Torque is transmitted without physical contact, minimizing friction and wear.

Magnet Design

- **Input:** 20 magnets on rotor side
- **Intermediate coupling**: 12 ferromagnetic nuts/bolts (act as flux guides)
- **Output:** 4 magnets on driven shaft

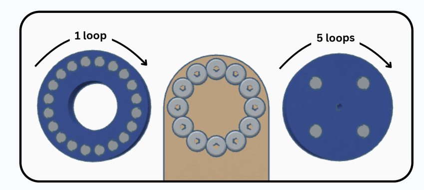
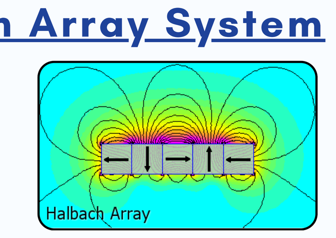


Image made by finalist using tinkercad

Gear ratio: 5:1 – output spins 5× faster than input

Dual Halbatch Array System

A Halbach array rotates magnetization vectors to concentrate magnetic flux on one face while canceling it on the opposite side.



What does a 2D Halbach array look like? (n.d.). Electrical Engineering Stack Exchange.

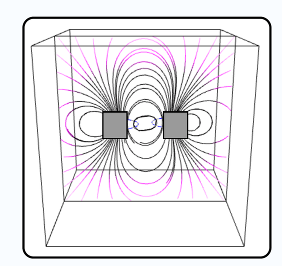


Image made by finalist using Falstad

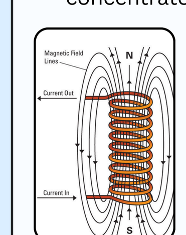
Two adjacent Halbach arrays create a localized magnetic flux minimum between them. This high-gradient region forms a passive radial restoring zone that stabilizes the rotor without mechanical contact.

Active Magnetic Bearings

Design Choices

- 45° angled electromagnetic coils enable bidirectional radial force control
- Coils wrapped around soft ferromagnetic poles to concentrate and amplify magnetic flux.


Image made by finalist using tinkercad


Electromagnets Explained: Types, Materials, Working Principles, and Key Applications.

Mechanism

- Current through coil generates magnetic field (Ampère's Law).
- Magnetic flux density scales with current and coil turns.
- Soft metal core amplifies force by increasing permeability (μ ≫ μ₀).

$$\oint \vec{B} \cdot \vec{dl} = \mu_o N I$$

Amperes law

Magnetic Thrust Bearing

The thrust bearing uses a Axial magnets resists forward/backward rotor displacement.

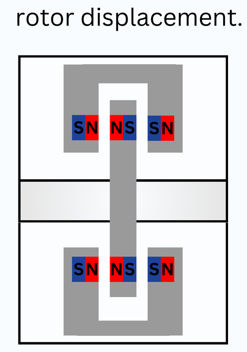
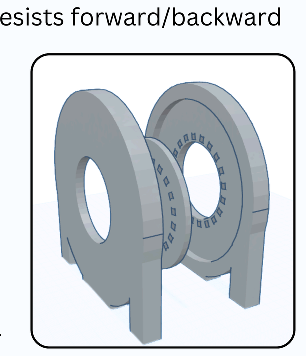


Image made by finalist using Canva

Mechanism

- As rotor shifts axially, flux density gradient increases restoring force.
- Acts as a passive equilibrium stabilizer for longitudinal motion.
- This elimates the issue of thrust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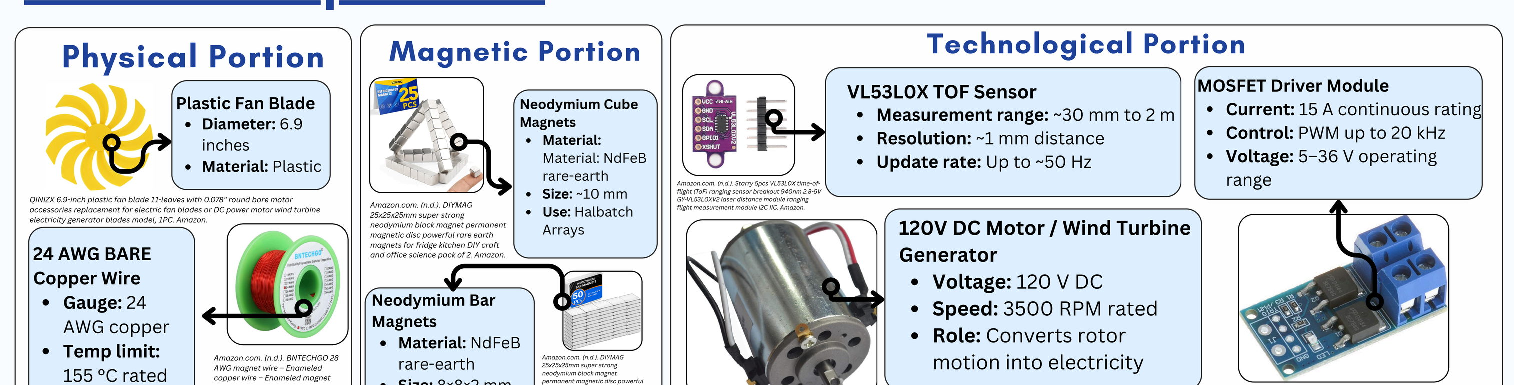
Other Components

Physical Portion



Plastic Fan Blade

- **Diameter:** 6.9 inches
- **Material:** Plastic

QINIZX 6.9-inch plastic fan blade 11-leaves with 0.078" round bore motor accessories replacement for electric fan blades or DC power motor wind turbine electricity generator blades model, 1PC. Amazon.

24 AWG BARE Copper Wire

- **Gauge:** 24 AWG copper
- **Temp limit:** 155 °C rated
- **Use:** AMB coils

Amazon.com. (n.d.). BNTECHGO 28 AWG magnet wire – Enameled copper wire – Enameled magnet winding wire – 4 oz – 0.0122" diameter 1 spool coil red temperature rating 155°C widely used for transformers inductors. Amazon.

Magnetic Portion

Amazon.com. (n.d.). DIYMAG 25x25x25mm super strong neodymium block magnet permanent magnetic disc powerful rare earth magnets for fridge kitchen DIY craft and office science pack of 2. Amazon.

Neodymium Cube Magnets

- **Material:** Material: NdFeB rare-earth
- **Size:** ~10 mm
- **Use:** Halbatch Arrays

Neodymium Bar Magnets

- **Material:** NdFeB rare-earth
- **Size:** 8×8×2 mm
- **Use:** Inside Rotor

Amazon.com. (n.d.). DIYMAG 25x25x25mm super strong neodymium block magnet permanent magnetic disc powerful rare earth magnets for fridge kitchen DIY craft and office science pack of 2. Amazon.

Technological Portion

Amazon.com. (n.d.). Starry 5pcs VL53L0X time-of-flight (ToF) ranging sensor breakout 940nm 2.8-5V GY-VL53L0XV2 laser distance module ranging flight measurement module I2C IIC. Amazon.

VL53L0X TOF Sensor

- **Measurement range:** ~30 mm to 2 m
- **Resolution:** ~1 mm distance
- **Update rate:** Up to ~50 Hz

MOSFET Driver Module

- **Current:** 15 A continuous rating
- **Control:** PWM up to 20 kHz
- **Voltage:** 5–36 V operating range

Amazon.com. (n.d.). 120V DC motor wind turbine generator power supply (3500rpm). Amazon.

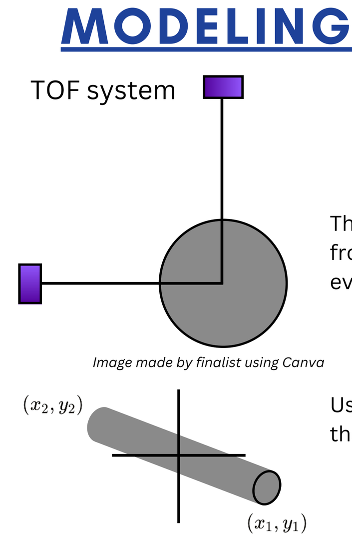
120V DC Motor / Wind Turbine Generator

- **Voltage:** 120 V DC
- **Speed:** 3500 RPM rated
- **Role:** Converts rotor motion into electricity

Amazon.com. (n.d.). MateIJS 11PCS MOSFET switch driver module, DC 5V-36V 15A (30A peak 400W) PWM control for motors & LED strips, low heat design, compatible with Arduino/Raspberry Pi. Amazon.

ROTOR CONTROL MECHANISM CALCULATIONS

MODELING THE ROTOR



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(x_1, y_1) (Sensor Plane 1)

(x_2, y_2) (Sensor Plane 2)

The X and Y data points are given to us from the TOF sensors, approximately every 0.01 seconds.

$$x_{CM} = \frac{x_1 + x_2}{2}, \quad y_{CM} = \frac{y_1 + y_2}{2}$$

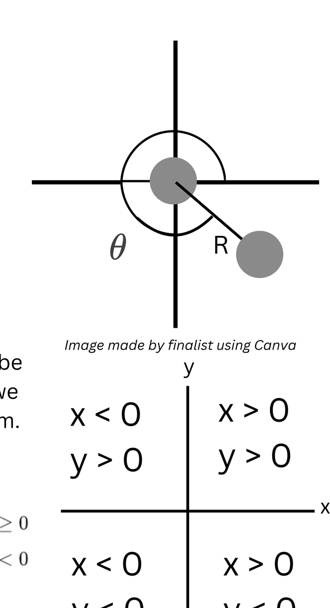
Using these data points, the location of the center of mass can be calculated.

Image made by finalist using Canva

$$x_{tilt} = \frac{x_2 - x_1}{2}, \quad y_{tilt} = \frac{y_2 - y_1}{2}$$

TOF system

$$R = \sqrt{x_{tilt}^2 + y_{tilt}^2}$$



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We can then use the tilt data from these points to calculate the angle and the magnitude of this tilt. The angle will also be depdent on the quadrant we're in. Thus we get our 3 ouputs for the prediction system.

$$\theta = \begin{cases} \arctan\left(\frac{y_{\text{tilt}}}{x_{\text{tilt}}}\right) & x_{\text{tilt}} > 0 \\ \arctan\left(\frac{y_{\text{tilt}}}{x_{\text{tilt}}}\right) + 180^\circ & x_{\text{tilt}} < 0 \text{ and } y_{\text{tilt}} \geq 0 \\ \arctan\left(\frac{y_{\text{tilt}}}{x_{\text{tilt}}}\right) - 180^\circ & x_{\text{tilt}} < 0 \text{ and } y_{\text{tilt}} < 0 \\ 90^\circ & x_{\text{tilt}} = 0, y_{\text{tilt}} > 0 \\ -90^\circ & x_{\text{tilt}} = 0, y_{\text{tilt}} < 0 \end{cases}$$

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PREDICTION OF MOVEMENT

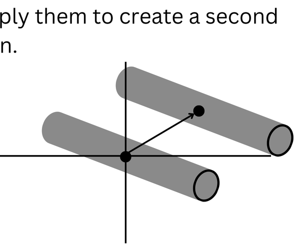
We continuously calculate the velocity and acceleration of each vector using a form of Euler's rule, and then reapply them to create a second order diffencial equation to predict motion.

$$v_x = \frac{x_{CM}(t) - x_{CM}(t-\Delta t)}{\Delta t}, \quad v_y = \frac{y_{CM}(t) - y_{CM}(t-\Delta t)}{\Delta t}$$

$$a_x = \frac{v_x(t) - v_x(t-\Delta t)}{\Delta t}, \quad a_y = \frac{v_y(t) - v_y(t-\Delta t)}{\Delta t}$$

$$x_{CM}^{future} = x_{CM} + v_x \Delta t + \frac{1}{2} a_x \Delta t^2$$

$$y_{CM}^{future} = y_{CM} + v_y \Delta t + \frac{1}{2} a_y \Delta t^2$$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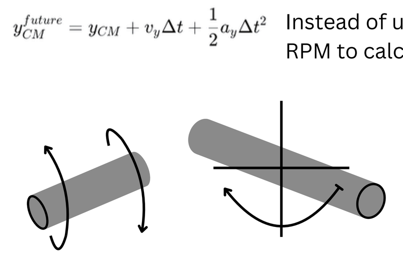
Instead of using eulars rule here, we used RPM to calculate angular motion instead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$$\omega_{\theta,RPM} = \frac{2\pi \cdot \text{RPM}}{60}$$

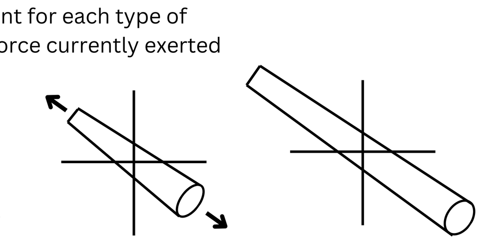
$$\alpha_\theta = \frac{\omega_\theta(t) - \omega_\theta(t-\Delta t)}{\Delta t}$$

$$\theta^{future} = \theta + \omega_\theta \Delta t + \frac{1}{2} \alpha_\theta \Delta t^2$$

This allowed us to account for each type of future motion, and the force currently exerted

$$v_R = \frac{R(t) - R(t-\Delta t)}{\Delta t}$$

$$a_R = \frac{v_R(t) - v_R(t-\Delta t)}{\Delta t}$$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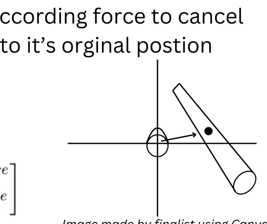
$$R^{future} = R + v_R \Delta t + \frac{1}{2} a_R \Delta t^2$$

OUTPUT OF VOLTAGE

Here we use Newtons laws to calculate the according force to cancel out the movment, then move the rotor back to it's orginal postion

$$\vec{F}_{CM} = -m\,\vec{a}_{CM} - k_{CM}\,\vec{x}_{CM}^{future} = \begin{bmatrix} F_{CM,x} \\ F_{CM,y} \end{bmatrix}$$

$$F_{tilt} = R^{future} k_\theta + m a_R \quad \vec{F}_{tilt} = F_{tilt} \begin{bmatrix} \cos \theta^{future} \\ \sin \theta^{future} \end{bmatrix}$$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We then combine the forces, and final the overal angle of direction of the force

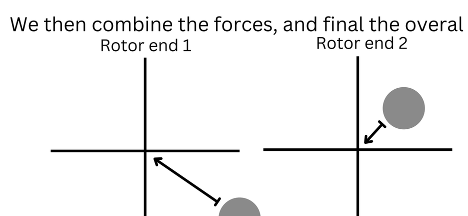


Image made by finalist using Canva

$$\vec{F}_{total} = \vec{F}_{CM} + \vec{F}_{tilt},$$

$$F_{total} = \sqrt{F_x^2 + F_y^2},$$

$$\theta_F = \text{atan2}(F_y, F_x)$$

Then we will distribute this force amoung the diffrent coils, which are each at a 45 degree angle within each diffrent quadrant.

Coils at $\phi_i = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

$$w_1 = \frac{\phi_2 - \theta_F}{90^\circ}, \quad w_2 = \frac{\theta_F - \phi_1}{90^\circ}, \quad w_1 + w_2 = 1$$

$$F_{coil1} = w_1 F_{total}, \quad F_{coil2} = w_2 F_{total}$$

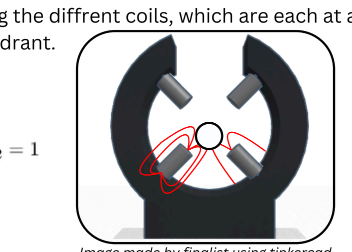


Image made by finalist using tinkercad

Finally we will turn this force into a specific amount of voltage, using a combination of Ampere's law and Ohm's law to translate. Here d represents distance, Rcoil is a constant based of coil design, and N is the number of loops

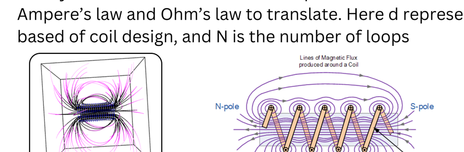


Image made by finalist using Falstad

Storr, W. (2018, February 9). Electromagnet, Electromagnetic Coil and Permeability. Basic Electronics Tutorials.

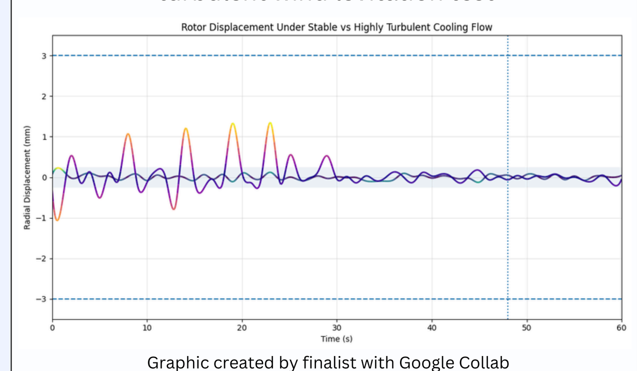
$$I_i = \frac{F_{coil,i}\, d_i^2}{N\, k_\mu}$$

$$V_i = R_{coil}\, I_i$$

This force pushes it back to its orginal postion, based off the softening variable k

TESTING & RESULTS

We have done a stable levitation test, and a turbulent wind levitation test



Graphic created by finalist with Google Collab

Test Condition	Max Displacement	Setting time	Peak Overshoot
Stable	0.15 mm	1.2 s	5%
Turbulent	1.17 mm	3.8 s	39%

Maximum Wind Speeds

327mph

This is the max speed that the thrust bearings can withstand.

Overall System Delay

1.37 Seconds

This is how long a full loop of sensing to voltage output would take.

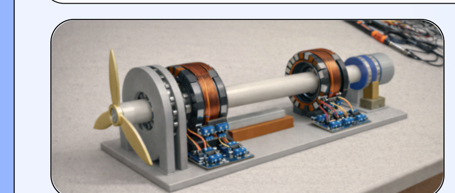


Image made By Finalist

The oscillation decreases as the RPM increases, clearly showing how the centripetal motion stabilizes the rotor

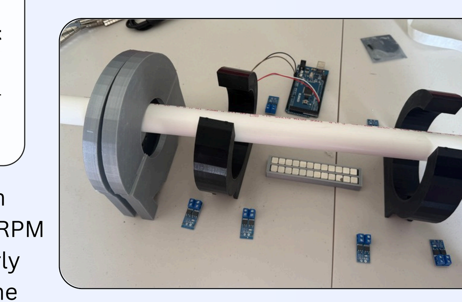


Image Taken by Finalist

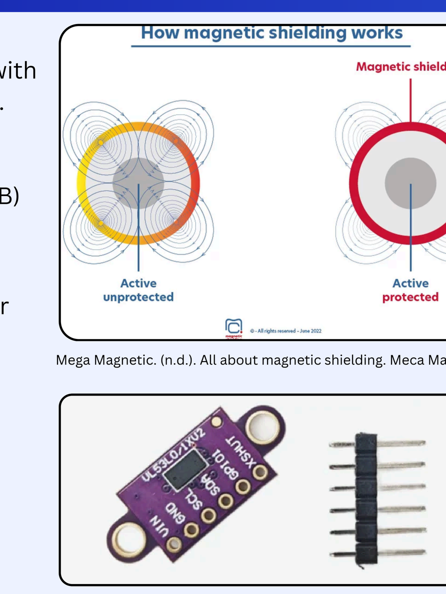
LIMITATIONS

Magnetic Field / Halbach Array

- **Limited lift distance:** Magnetic field decays rapidly with distance; small air gap misalignments can reduce lift.
- **Stray flux:** Some leakage can interfere with nearby electronics
- **Temperature sensitivity:** Permanent magnets (NdFeB) lose strength at high temperatures

Active Magnetic Bearings (Coils + Sensors)

- **Power consumption**: Coils require continuous power for active control; high loads or disturbances increase energy usage.
- **Sensor and control limits:** ToF sensors and microcontroller have finite response times; high-speed rotor or sudden disturbances may exceed correction speed.
- **Complex tuning required**: PID or feedback parameters must be carefully tuned to prevent oscillations or instability.



Mega Magnetic. (n.d.). All about magnetic shielding. Meca Magnetic.

Amazon.com. (n.d.). Starry 5pcs VL53L0X time-of-flight (ToF) ranging sensor breakout 940nm 2.8-5V GY-VL53L0XV2 laser distance module ranging flight measurement module I2C IIC. Amazon.

CONCLUSIONS

We successfully demonstrated a hybrid magnetic levitation system combining permanent magnets and actively controlled electromagnetic coils. We achieved near-contactless rotor support, significantly reducing mechanical friction compared to traditional bearings. Thus have proved the feasibility and scalability of magnetically supported wind turbine rotors.

Key Challenges

- **Eddy current drag:** Induced currents in nearby conductive materials created unwanted drag
- **Magnetic interference:** Interaction between control coils and permanent magnets complicate stabilization

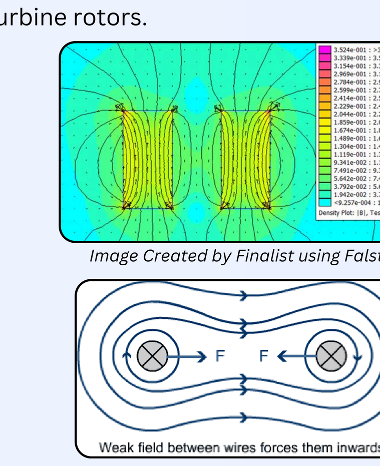


Image Created by Finalist using Falstad

Overall Improvements

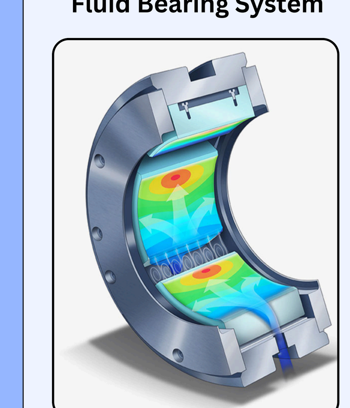
- **Less energy usage:** Developed a hybrid passive–active magnetically levitated, using other magnetic systems to fully remove friction
- **No wear**: Extends operational lifespan while lowering maintenance requirements
- **Horizontal Maglev Turbine:** First Designed, allowing for higher efficiency and scalability for industrial turbine

Thesat, S. (2009, April 14). Levitation and diamagnetism, or: LEAVE EARNSHAW ALONE!!! Skulls in the Stars.

FUTURE EXTENTIONS & ALTERNATIVES

Fluid Bearing System

These use a layer of oil to reduce friction between moving parts. Not used because it still causes friction, wear, and energy loss at high speeds.


Three Fluid Film Journal Bearing Models from Kingsbury. (2025). Kingsbury.com.


Electricalvolt. (2022, October 18). LASER Sensors Explained-Types, Working Principle & Advantages. Electrical Volt.

Laser Sensors

We could use lasers to measure rotor position very precisely. They were not used because they are expensive and can be affected by dust or vibration.


Vicinanza, D. (2025, February 10). AI is transforming the search for new materials that can help create the technologies of the future. The Conversation.

Superconductivity for Repulsion

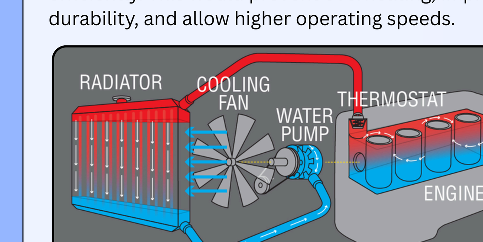
Uses superconducting magnetic systems to levitate the rotor. This was not used because it requires extreme cooling and is too costly for large systems.

Smart materials & Cooling System

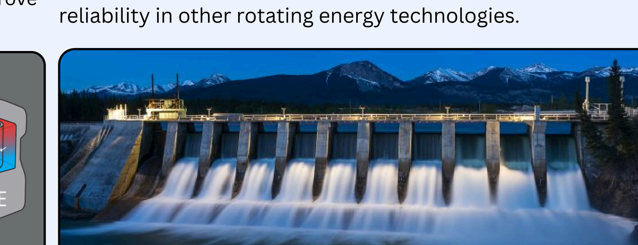
A closed-loop water cooling system could remove heat from the coils and power electronics more efficiently. This would prevent overheating, improve durability, and allow higher operating speeds.

Application in other Fields

This magnetic bearing system could also be used in tidal turbines, hydro systems, or flywheel storage to improve efficiency and reliability in other rotating energy technologies.



byfinshap. (2021, August 26). How Automotive Cooling Systems Work - BG Find A Shop. Bgfindashop.com.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